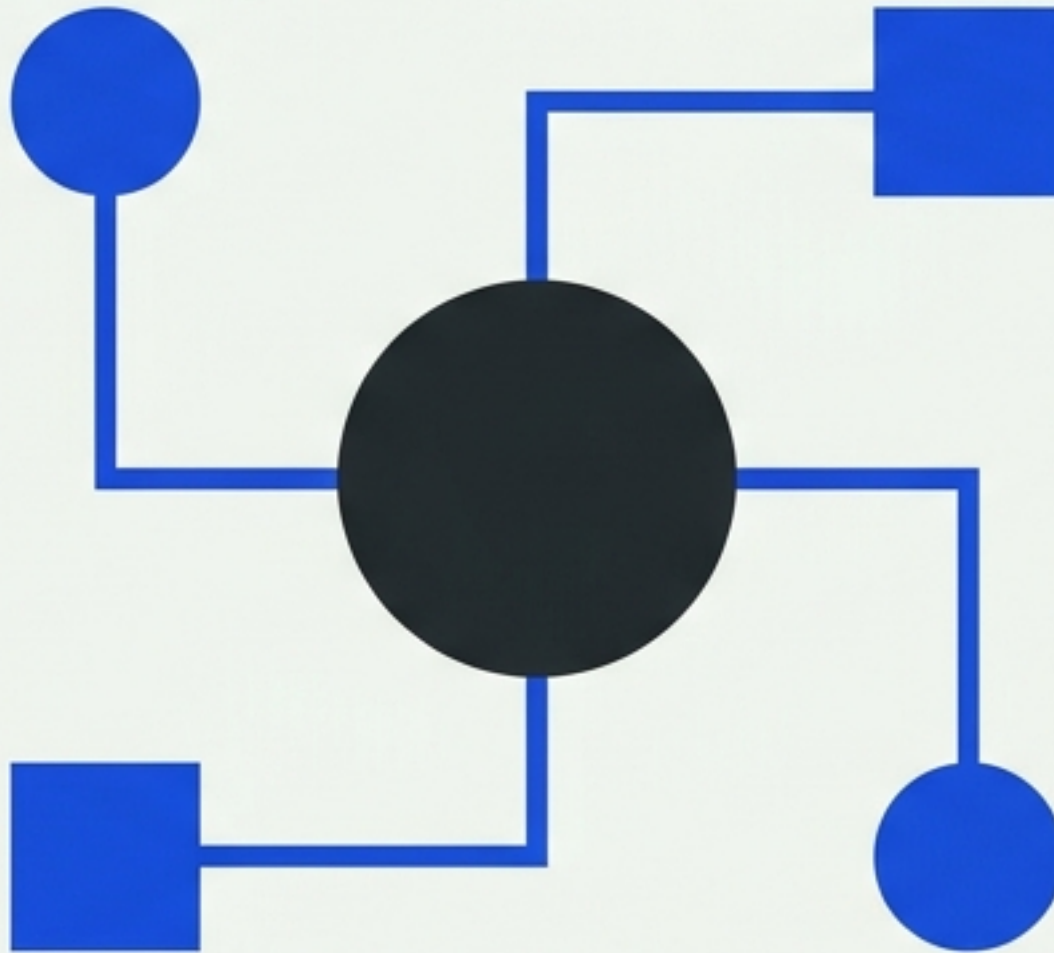


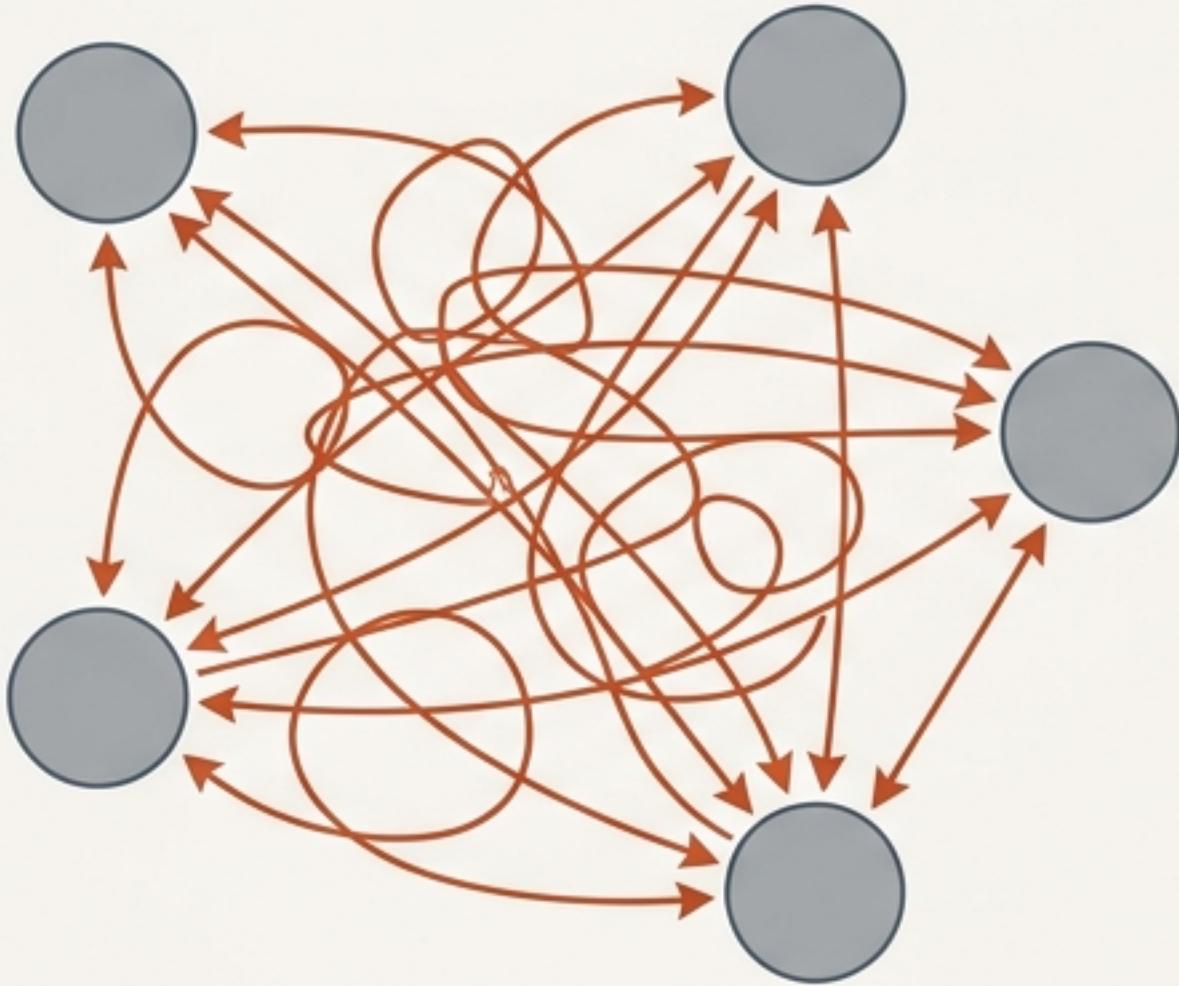
The Event-Driven War Room

A real-time, multi-agent incident response system built on asynchronous event streams.



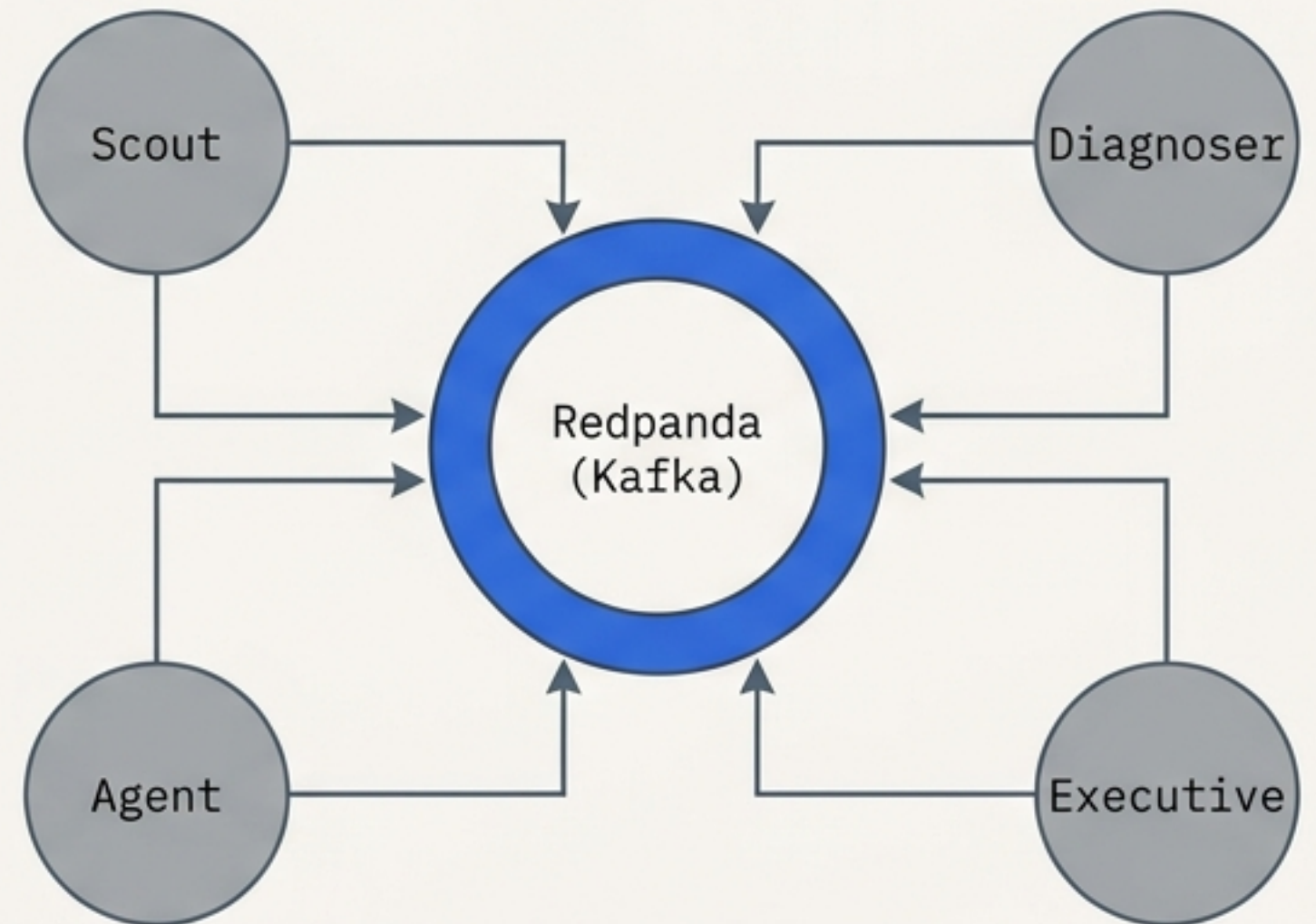
Moving Beyond Synchronous 1-to-1 Chat

The Old Way: Synchronous API Chains



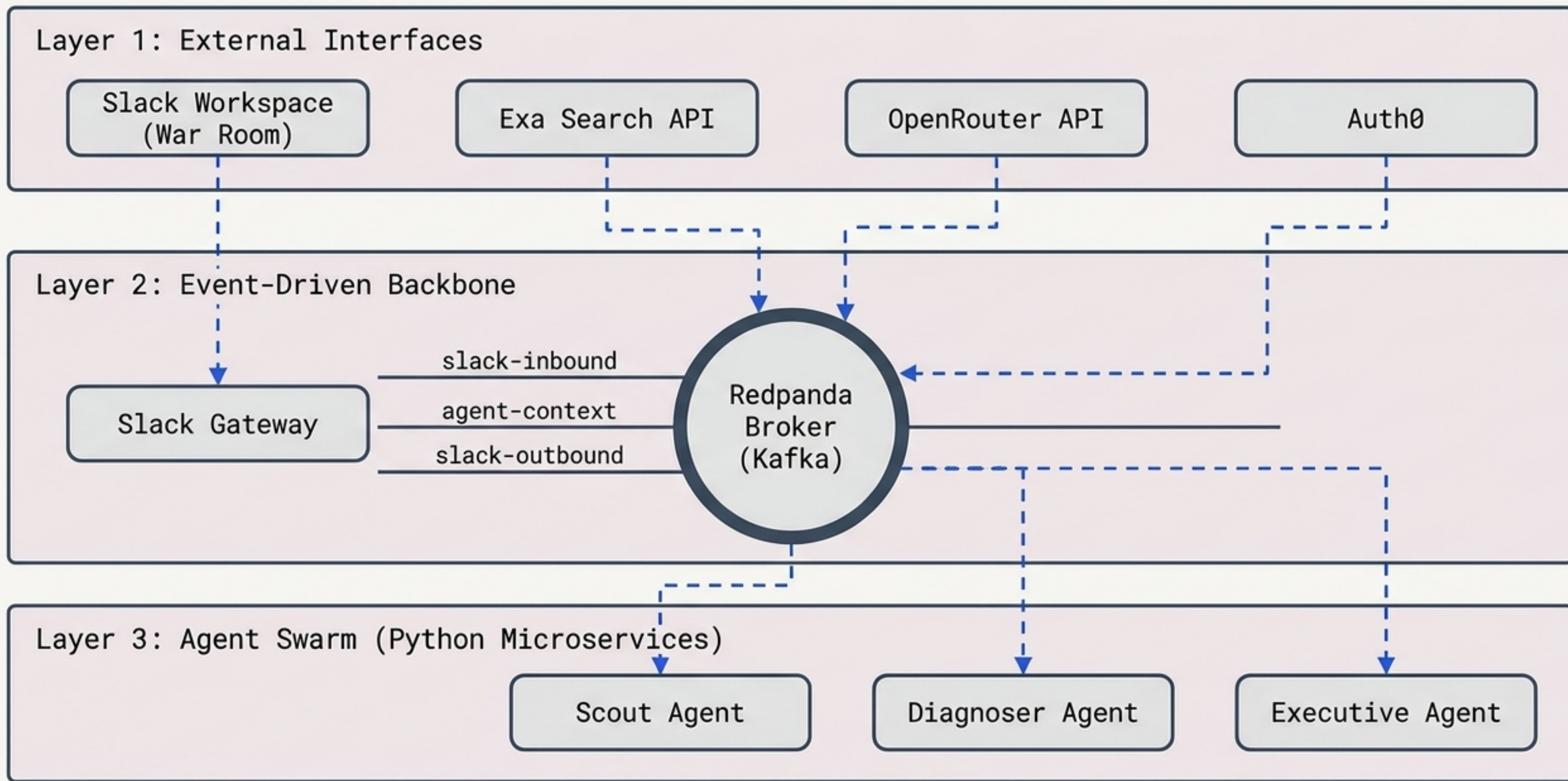
Rigid, blocking requests where agents wait on each other, creating bottlenecks.

The New Paradigm: Asynchronous Event Streams



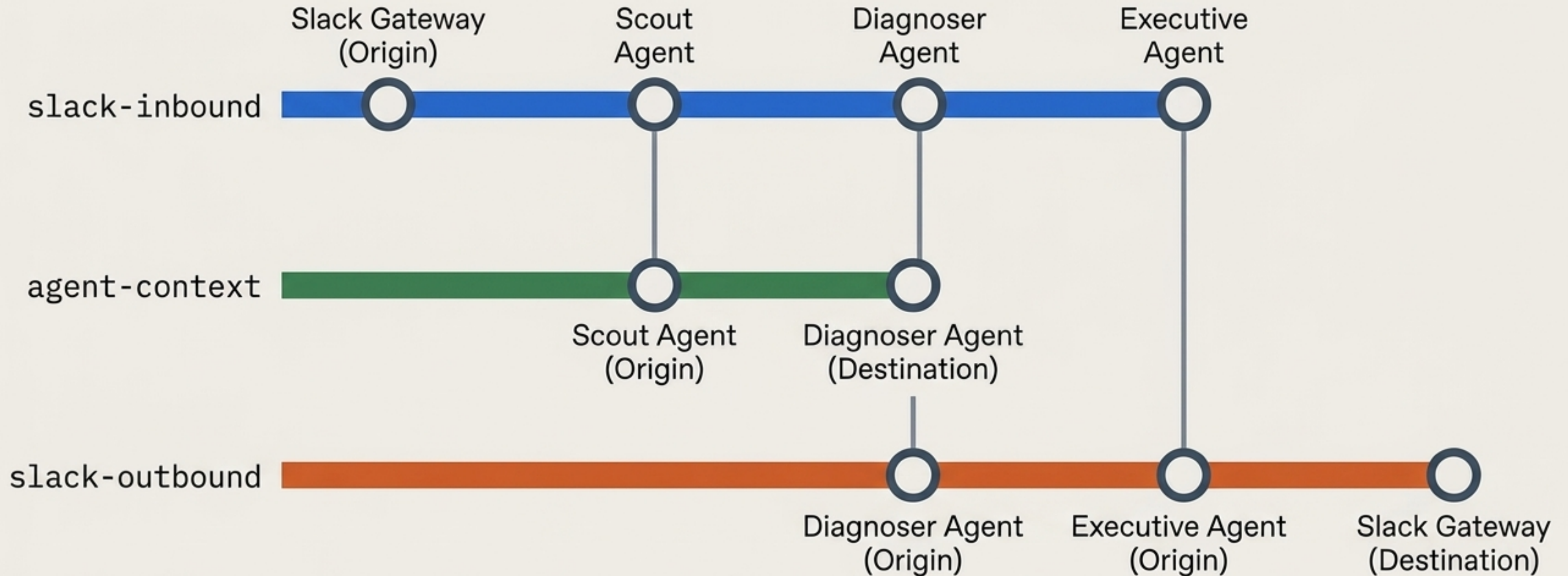
Agents operate asynchronously and in parallel, reading and writing to a central environment independently.

Mapping the Blackboard Architecture



Tracing the Event-Driven Backbone

Abstract Kafka topics transformed into tangible data transit routes carrying events between decoupled system components.

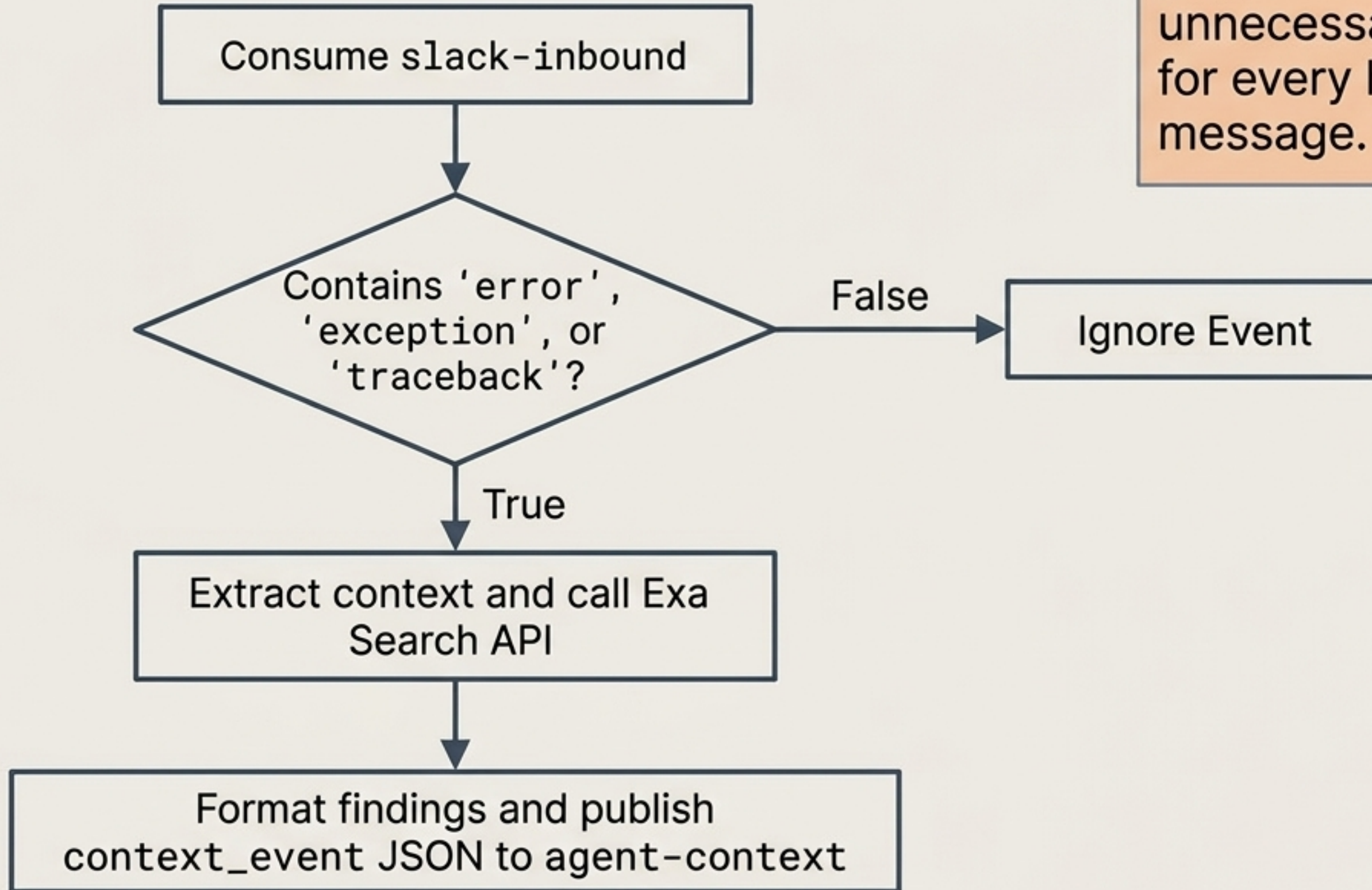


The Agent Swarm Diagnostic Matrix

	Scout (The Researcher)	Diagnoser (The Brain)	Executive (The Actor)
Trigger Event	Consumes slack-inbound (looks for error traces)	Consumes agent-context & slack-inbound	Consumes slack-inbound (looks for commands)
External Tooling	Exa Search API	OpenRouter (Claude 3.5 Sonnet)	Auth0 & Rollback API
Output Destination	Silently publishes to `agent-context`	Synthesizes to `slack-outbound`	Mutates state and publishes to `slack-outbound`

Scout Agent Logic and Ambient Research

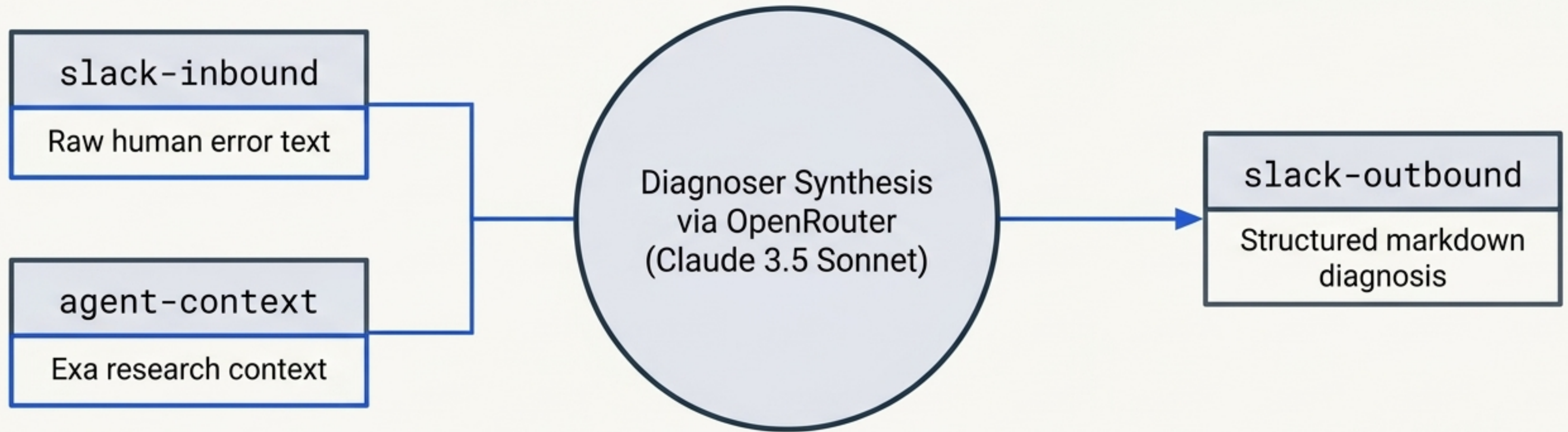
Lightweight heuristics prevent expensive, unnecessary API calls for every human chat message.



Diagnoser Agent

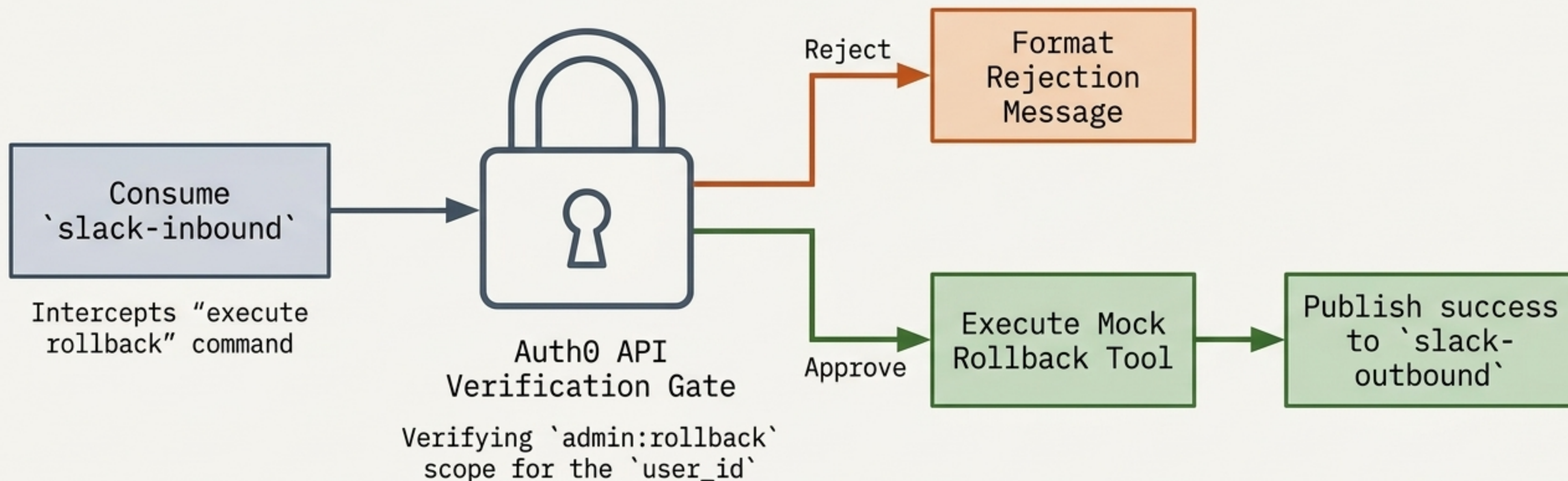
Synthesis and Root Cause Analysis

Ingestion Funnel



The Diagnoser does zero web searching itself. It acts strictly as the brain, merging human input with the Scout's automated research asynchronously.

Executive Agent and Strict Authorization Gates



Because this agent mutates state (reverting deployments), it requires explicit human commands, bypassing ambient listening entirely.

Event Stream Contracts Define the System

Agents have no awareness of who consumes their data; they strictly trust the shape of the JSON payload.

Topic: slack-inbound

```
{
  "channel_id": "C12345678",
  "user_id": "U98765432",
  "text": "...Auth0
unauthorized audience
error..."
}
```

Topic: agent-context

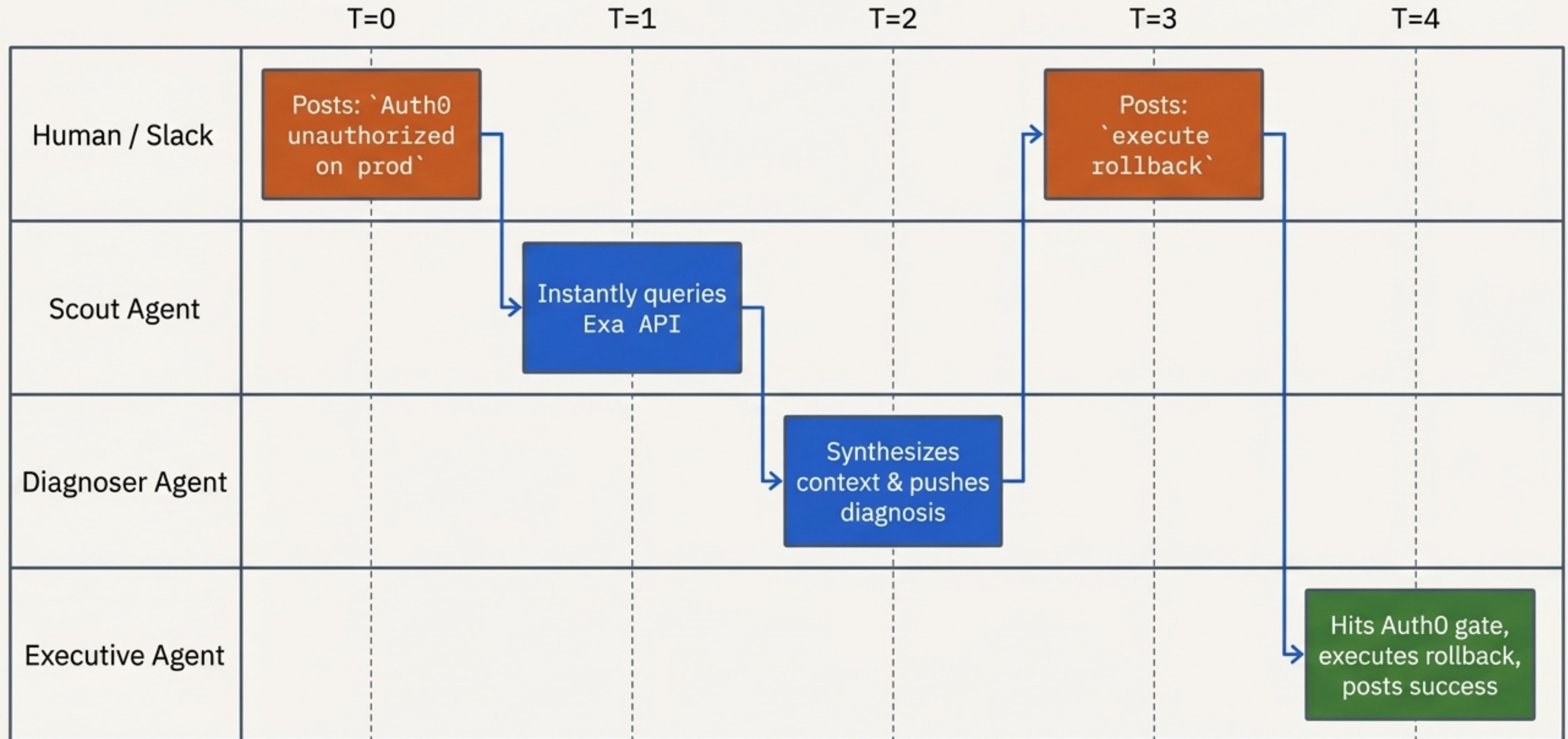
```
{
  "original_text":
"...unauthorized audience
error...",
  "research_context": "Exa
Search Result...",
  "source": "exa_scout"
}
```

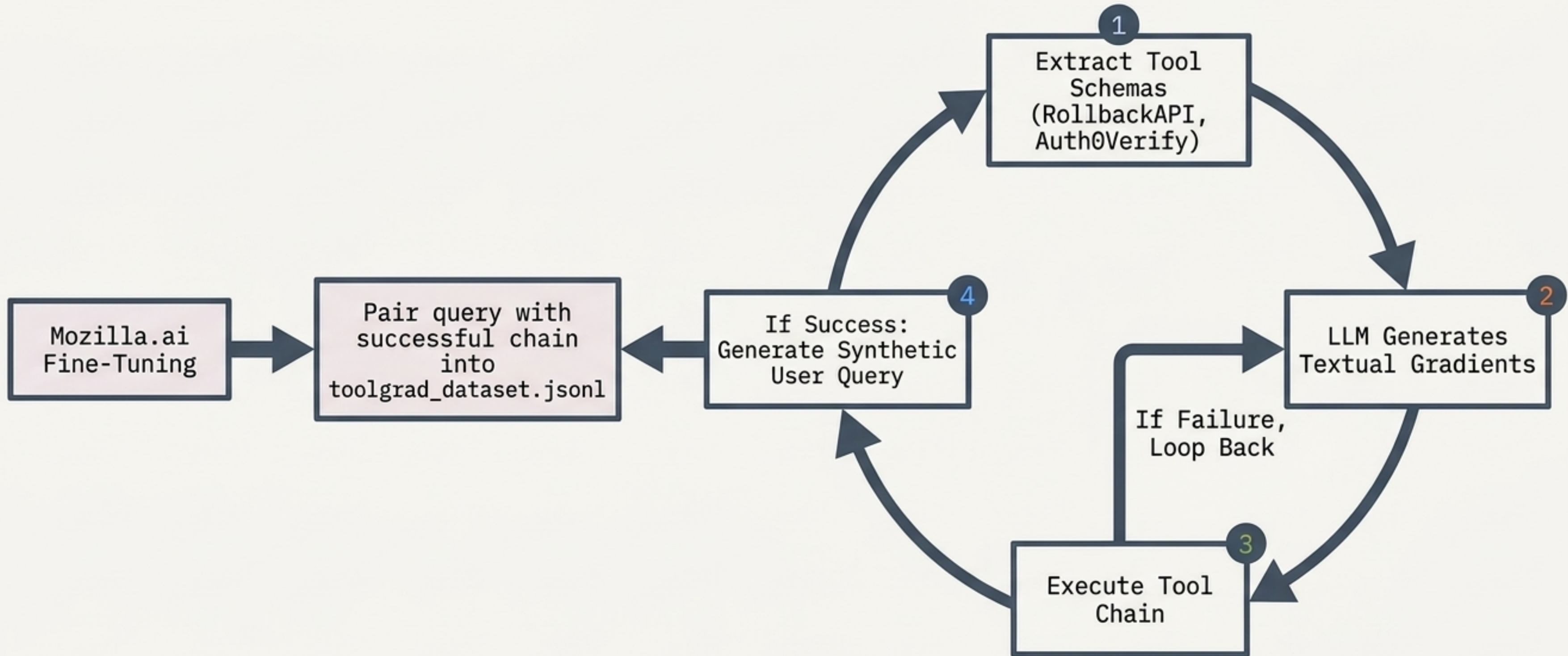
Topic: slack-outbound

```
{
  "channel_id": "C12345678",
  "text": ":white_check_mark:
Executive Action Confirmed...
Rollback successful."
}
```


Anatomy of a Sev-1 Resolution

Zero explicit @mentions required.





This guarantees 100% verified tool execution in the training data before the Executive Agent is unleashed in the War Room.

The Event-Driven War Room Tech Stack

